

Santo Gezelin

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PROFILE

Mechanical engineer with a robotics minor and graduate coursework in robot control, motion planning, and perception, now working in field applications and integration: acceptance testing, commissioning documentation, and hardware triage on installed systems. Savannah based and available for multi shift coverage without relocation.

EDUCATION

B.S. Mechanical Engineering, Robotics Minor | University of Utah

May 2026

GPA 3.65 | Dean's List, seven semesters | WUE Scholarship | Graduate level coursework in control, motion planning, manipulation, perception

TECHNICAL SKILLS

Programming and Simulation

- Python, C++, MATLAB
- Simulink, PyBullet, ROS
- Embedded C, Bash, Git

Artificial Intelligence

- Local LLM deployment (Ollama)
- LLM API integration, tool calling
- Agentic workflow design

Systems and Tools

- Linux
- Bluebeam, Microsoft Office

Autonomy and Control

- Motion planning (A*, RRT*, Dijkstra, D* Lite)
- Receding horizon planning
- Monte Carlo reachable set sampling
- PID and state space control
- Hybrid position and force control
- Impedance and admittance control
- Operational space and sliding mode control
- Visual servoing (IBVS)

Mechanical and CAD

- SOLIDWORKS (CSWA certified)
- AutoCAD, Autodesk Inventor
- 3D printing, laser cutting, machining

Test and Instrumentation

- Factory acceptance testing (FAT)
- Root cause and fault isolation
- Instrumentation and control document review
- Sensor and telemetry integration
- Thermal validation, continuity testing

Electronics and Hardware

- PCB design, soldering and wiring
- H bridge drivers, shift registers
- Arduino (Mega, Uno), XBee

PROFESSIONAL EXPERIENCE

Staff Mechanical Engineer | Beacon Water Technologies / Orthos Liquid Systems

Aug 2026 to Present

Savannah, Georgia | Pile cloth media filtration for municipal water reclamation

- Preconstruction engineering: sizing and configuration of pile cloth media filtration systems
- Development of proposals, competitive bids, and complete equipment submittal packages
- Submittal ownership: assembly, tracking, and revision through engineer of record approval
- Submittal accuracy and completeness checked against specifications, drawings, and schedule
- Instrumentation and control submittal review: level transmitters, float switches, analyzers, drives, panels
- Tag discrepancy and open RFI resolution direct with the design engineer of record
- Vendor and procurement coordination, plus onsite receiving and inspection against approved submittals

Data Systems Engineer Intern | Quantum Surgery Center

May 2025 to Aug 2025

Newport Beach, California

- Electronic medical record data cleaning and restructuring: 40 percent improvement in query reliability
- Duplicate record reduction and reorganization to enable reliable transfer between systems
- Tracking system for 500 plus records across intake, consultation, and conversion stages

Research and Development Engineer Intern | BrightUro

Jun 2024 to Aug 2024

Irvine, California | Medical device startup

- Root cause analysis to localize failures to hardware versus software
- Subsystem level fault isolation before escalation to the responsible engineer
- Robotic assembly solution designed, prototyped, and deployed independently in a startup with no existing automation
- Human factors testing protocols supporting an FDA 510(k) regulatory submission
- 3D printed fixtures and tooling in SOLIDWORKS to reduce assembly time and operator variability

ENGINEERING PROJECTS

Autonomous Aerial Interception | Graduate Motion Planning

Jan 2026 to May 2026

- Receding horizon motion planner in Python for real time interception of a maneuvering target, with Monte Carlo reachable set sampling to predict stochastic target motion
- Defined success criteria and validation campaign: 85.8 percent interception over 500 Monte Carlo trials, visualized in PyBullet

512 Pixel Kinetic Flip Disc Installation | Senior Capstone

Sep 2025 to May 2026

Permanent installation, J.W. Marriott Jr. Institute building, University of Utah

- Led electromechanical design from custom PCB through permanent commissioned install; H bridge drivers and shift registers so the array scaled modularly
- Hand wound and tested 559 electromagnets at 91.6 percent yield in band; corrected the winding procedure mid build from the failure distribution
- Thermal validation via resistance-based temperature monitoring; roughly 8,400 soldered joints with continuity and fault isolation testing

Competition Mechatronics Robot | Software Lead

2025

- Rebuilt the full robot codebase from scratch in 48 hours after the control software failed during integration, and competed on schedule
- Arduino Mega and Uno for motor control, sensing, and XBee links; PID control for line following and motor positioning